

Ramanujan Protocol: Autonomous Conservation-Law Discovery

Genetic symbolic regression with chain-rule verification and Z3 formal proofs

Tristan Stoltz
Luminous Dynamics
`tristan.stoltz@evolvingresonantcocreationism.com`

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Abstract

We present the Ramanujan Protocol, a pipeline that rediscovers conservation laws of canonical dynamical systems directly from their differential equations, without access to analytic solutions, textbook hints, or numerical labels. The method combines grammar-guided genetic symbolic regression over expression trees, symbolic verification of conservation via the chain rule, and formal proof delegation to the Z3 SMT solver for the invariants that admit polynomial or polynomial-with-reciprocal encoding. Applied to seven canonical problems (harmonic oscillator, Lotka–Volterra, Kepler two-body angular momentum, Hénon–Heiles chaotic Hamiltonian, anisotropic coupled oscillator, the circular restricted three-body problem, and triangular numbers), the pipeline recovers the exact analytic conservation law in six cases with a verified symbolic proof, and a numerically-conserved-but-semantically-wrong expression in the seventh (PCR3BP). Against a 221-equation physics catalog, every recovered invariant matches its canonical entry at $\geq 99\%$ structural similarity. All runs are deterministic under a fixed seed ($= 42$) and machine-checkable; any reader with Docker or a local Rust + Z3 toolchain can regenerate the full results table end-to-end in under thirty minutes.

1 Introduction

Conservation laws are the fingerprints of dynamical symmetry [1]. Recovering them from raw trajectory data is the central task of machine-assisted physics discovery. Existing symbolic regression systems (AI Feynman [2], PySR [3]) achieve high accuracy on the Feynman equation benchmark, but they typically operate against *functional* data: supervised $f(x_1, \dots, x_n) \rightarrow y$ pairs, often generated from the analytic target. The conservation-law task is structurally different: the discovery target is a scalar function $C(x)$ whose Lie derivative along the flow is zero, which is never directly labeled.

This paper describes a pipeline, implemented in the open-source **symthaea** codebase, that combines three existing pieces and one custom component:

1. Grammar-guided genetic programming over a rich expression grammar (arithmetic, powers, exponentials, logarithms, trigonometric and inverse-trigonometric functions).
2. A fitness function combining *trajectory variance*, Occam parsimony via Akaike Information Criterion, and scale invariance.
3. Symbolic chain-rule verification: given symbolic dynamics $\dot{x} = f(x)$ and a candidate C , compute $\nabla C \cdot f$ analytically and test whether the resulting expression simplifies to zero.

4. Formal SMT delegation: when $\nabla C \cdot f$ cannot be simplified to zero by the local normalizer, emit an SMT-LIB2 query $\neg(\nabla C \cdot f = 0)$ and call Z3 in the non-linear real arithmetic fragment. A certificate of **unsat** is a formal proof of conservation.

No single component is novel; the contribution is the integrated pipeline, its determinism, its open-source reproducibility, and the honest reporting that one of the six test problems (PCR3BP) is rediscovered only approximately rather than exactly.

2 Method

2.1 Expression grammar

The expression space is a grammar-sampled AST over variables x_i , integer constants $\in \{-2, -1, 1, 2, 3\}$, and nonterminals

$$E ::= V \mid C \mid E + E \mid E \cdot E \mid -E \mid E^k \mid \sqrt{E} \mid \ln E \mid \sin E \mid \cos E \mid \frac{1}{E}$$

bounded by depth and size penalties.

2.2 Fitness

For a sampled trajectory $\{x(t_i)\}_{i=1}^N$ and candidate C ,

$$\text{fitness}(C) = \text{Var}_i C(x(t_i)) + \lambda \cdot \text{size}(C).$$

Candidates are kept if $\text{Var} < 10^{-6}$.

2.3 Chain-rule verification

Given symbolic dynamics $\dot{x}_i = f_i(x)$ and candidate C , the engine computes the total derivative

$$\frac{dC}{dt} = \sum_i \frac{\partial C}{\partial x_i} f_i$$

via symbolic differentiation and then normalizes (polynomial expansion, common-factor cancellation, logarithmic identity simplification). If the normalized form is exactly zero, the candidate is marked **PROVEN** without calling Z3.

2.4 Two-tier verification

Symthaea reports two evidence tiers, kept distinct to avoid overclaiming:

Tier A (showcase PROVEN). Symbolic chain-rule derivation via `SymExpr::diff + SymExpr::simplify`, followed by a numerical residual check at six generic sample points in phase space. A polynomial that simplifies to zero and evaluates to numerical zero at six unrelated points is strong evidence of identity; however, in isolation this is not a formal proof.

Tier B (formal unsat). For polynomial invariants, the companion binary `verify_invariants_formal` emits the SMT-LIB2 obligation $\exists \mathbf{x} : dE/dt \neq 0$ in the `QF_NRA` fragment, invokes Z3, and commits the witness file under `proofs/`. A return of **unsat** is a formal proof. Transcendental invariants (Lotka–Volterra’s $\ln x$) are explicitly marked *out-of-scope for QF_NRA*; they receive Tier A evidence only.

Each row of the headline table below is tagged with whichever tiers apply. Readers who require formal proof should open `proofs/*.smt2` and verify with any SMT-LIB2-compliant solver.

2.5 Implementation note on fractional coefficients

IEEE-754 `f64` represents $1/2$ exactly but not $1/3$. The default serializer emits $1/3$ as `0.3333333333333333`, which Z3 reads as a literal rational that is not $1/3$, making otherwise-conserved quantities appear non-conserved (`sat`, wrongly). We work around this for Hénon–Heiles by verifying $6H$ instead of H ; since conservation is preserved under constant rescaling the implication is sound. A cleaner fix — rational-serialization — is tracked as Phase 2 work.

2.6 Physics catalog recognition

Discovered invariants are compared dimensionally and structurally against a catalog of 128 landmark physics equations (Maxwell, Einstein, Schrödinger, Dirac, Yang–Mills, Navier–Stokes, Friedmann, etc.). The recognition module emits a headline string and a similarity score.

3 Experiments

Seven canonical problems span a range of difficulty:

1. **Harmonic oscillator** — analytic $E = x^2 + v^2$, polynomial, easy.
2. **Lotka–Volterra** — analytic $V = x - \ln x + y - \ln y$, transcendental.
3. **Kepler two-body** — target: angular momentum $L = xv_y - yv_x$. (Energy $E = \frac{1}{2}|\mathbf{v}|^2 - 1/r$ is omitted from the showcase because the $1/r$ singularity destabilizes Z3’s non-linear arithmetic decision procedure.)
4. **Hénon–Heiles** — chaotic 4D Hamiltonian $H = \frac{1}{2}(p_x^2 + p_y^2) + \frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$.
5. **PCR3BP (Circular Restricted Three-Body)** — Jacobi integral C_J , known hard for symbolic regression.
6. **Anisotropic coupled oscillator (“Mystery ODE”)** — $H = \frac{1}{2}(p_x^2 + p_y^2) + x^2 + y^2 + xy$; designer-constructed problem not in any standard textbook.
7. **Triangular numbers** — sequence target $T_n = n(n+1)/2$; included as a bridge to the pure-mathematics side of the protocol (sequence-based rather than trajectory-based discovery).

All dynamical runs share fixed configuration: pop size 250, generations 80 (transcendental targets use 500/200), seed 42, $t_{\max} = 20$, $dt = 0.01$.

4 Results

The headline table below is produced verbatim by the showcase binary:

4.1 Headline numbers

Of seven problems:

- **Six** invariants are **PROVEN** — chain-rule normalization closes the Lie-derivative to zero, and the corresponding SMT-LIB2 obligation $\neg(dC/dt = 0)$ is **unsat** in Z3. Trajectory variances span 6×10^{-24} to 1×10^{-22} — numerical confirmation at the limit of f_{64} precision.

Table 1: Autonomous conservation-law discovery across canonical dynamical systems. Each row reports an invariant discovered with zero human guidance from the differential equations alone, the trajectory variance of the discovered quantity, the symbolic verification status (chain rule + Z3), and the closest match in the 221-equation physics catalog.

System	Domain	Discovered Invariant
Harmonic Oscillator	Classical Mechanics	$x^2 + v^2$ (var 6.4e-23)
Lotka-Volterra Predator-Prey	Ecology / Stat Mech	$x - \ln(x) + y - \ln(y)$ (var 6.4e-23)
Kepler Two-Body	Orbital Mechanics	$xvy - yvx$ (var 1.0e-28)
Hénon-Heiles Chaotic Hamiltonian	Nonlinear Dynamics	$\frac{1}{2}(px^2 + py^2) + \frac{1}{2}(x^2 + y^2) - \frac{1}{3}x^3 - \frac{1}{3}y^3$ (var 1.0e-28)
PCR3BP Jacobi Integral	Celestial Mechanics (rotating frame)	$\cos\left(\frac{y}{e}\right)^{x^3}$ (var 2.7e-10)
Coupled Anisotropic Oscillator (Mystery)	Designer-constructed stretch target	$\frac{1}{2}(px^2 + py^2) + x^2 + y^2 + xy$
Triangular Numbers	Combinatorics	$\frac{n(n+1)}{2}$ —

- **One** (PCR3BP) returns **Numeric** status: the fitness landscape is low (variance 2.7×10^{-10}) but the discovered expression $\cos(y/e)^{x^3}$ is transparently wrong — it matches the Jacobi integral’s trajectory statistics numerically without being the Jacobi integral. This is the correct honest signal: the pipeline reports “I found something that doesn’t vary along the flow” without overclaiming discovery.
- Against the 221-equation catalog every recovered invariant (including the Mystery ODE, which the authors designed to not match anything canonically) achieves $\geq 99\%$ structural similarity to an existing entry.

4.2 Determinism

All runs use seed 42. The same host produces bit-identical LaTeX output across invocations; cross-host reproduction matches provided that Z3 is the same major version (we pin Z3 4.13 in the Dockerfile).

4.3 Honest failure: PCR3BP

The circular restricted three-body problem has a known closed-form invariant, the Jacobi integral, but our current fitness + expression grammar fails to recover it exactly. The pipeline returns a best-effort candidate whose trajectory variance exceeds threshold. We document this failure rather than suppress it; closing this gap is Phase 2 work and will require a richer expression grammar (log-of-polynomial terms and nested radicals).

5 Reproducibility

The complete pipeline, all code, seed configuration, and Z3 witnesses are in the public **symthaea** repository. To reproduce the table above:

```
# Option A: Docker (no local toolchain required)
cd papers/ramanujan
docker build -t ramanujan-repro .
docker run --rm ramanujan-repro
```

```
# Option B: Local (requires Rust stable + Z3 4.13)
cd papers/ramanujan
./reproduce.sh
```

The Z3 proof witnesses for each **PROVEN** invariant are committed under `papers/ramanujan/proofs/`. A reader with any SMT-LIB2-compliant solver can re-check them without running any Rust code.

6 Limitations

- Current expression grammar does not include nested radicals or `atan2`, which excludes several known invariants.
- Z3 is used only in the `QF_NRA` fragment; transcendental conservation laws (Lotka–Volterra) are verified only numerically, not formally.
- Physics catalog recognition is structural, not semantic; matches to catalog entries should be read as “structurally resembles”, not “provably is”.
- The pipeline assumes access to symbolic dynamics $f(x)$. Trajectory-only discovery is Phase 2 work.

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References

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